

TAMS Tower Suitability — Complete Scoring Reference

Transmission Asset Intelligence Platform · Open-data screening only · Generated for TAMS Tower Suitability module · Not a design certificate

A. Final Output

Output	How it is calculated
Site score (0–10)	Weighted average of 12 factors (each scored 0–10 first). Formula: $\Sigma(\text{factor_score} \times \text{weight}) \div \Sigma \text{weights}$
Verdict	≥ 7.0 Preferred · 4.5–7.0 Conditional · < 4.5 Unsuitable (screening)
Confidence %	Base 55% + up to 25% for resolved live signals – 5% per Photon fallback used

Confidence sub-points (7 signals)

- Elevation resolved · Slope resolved · Road distance · Water distance · Settlement distance · Grid (tower/SS) · Wind

Fallback penalties (–5% each)

- Water Photon fallback · Settlement Photon fallback · Grid data unavailable fallback

B. Scoring Factors (12 total)

1. Terrain slope Weight 22%

Sub-point	Detail
DEM source	Open-Meteo Elevation API → Copernicus DEM ~30 m
Sample pattern	5 points: centre pad + N/S/E/W at ~133 m offset (0.0012°)
Slope calculation	Max angle (°) from centre to each offset: $\text{atan}(\Delta\text{elev} / \text{run}) \times 180/\pi$
Output	Maximum slope among 4 directions

Slope	Score
$\leq 5^\circ$	10 (best)
$\geq 18^\circ$	0 (worst)
Between	Linear interpolation

2. Road access Weight 14%

Sub-point	Detail
API	OSRM Nearest (router.project-osrm.org/nearest/v1/driving)
Road network	OpenStreetMap drivable roads
Distance	OSRM snap distance (m) → km; Haversine fallback
Map output	Nearest road snap lat/lon (orange marker)

Road distance	Score
≤ 0.5 km	10
≥ 8 km	0
Lookup failed	3 (uncertain)

3. Water / flood buffer Weight 14%

OSM Overpass (primary) — 8 km radius

Feature	OSM tag
Natural water	way/relation natural=water
Waterways	way waterway
Reservoirs / basins	landuse=reservoir, landuse=basin
Generic water	way water

Photon fallback (if Overpass fails)

Query: "lake river reservoir" · osm_tag=natural:water · –5% confidence

Water distance	Score logic
≥ 0.8 km from water	10 (far = good)
≤ 0.05 km (~50 m)	0 (very close)
< 0.15 km	Flood/scour screening flag

4. Settlement clearance Weight 10%

OSM Overpass — 4 km radius

Feature	OSM tag
Buildings	way building
Places	node place=city town village hamlet suburb
Residential	way landuse=residential

Settlement distance	Score
≥ 0.25 km	10
≤ 0.02 km	0
< 0.08 km	ROW/social risk flag

5. Power connectivity Weight 10%

Sources (radius 8 / 15 / 25 / 50 km)

Source	Assets
TAMS GIS	Towers, substations, transmission lines
OSM Overpass	tower, portal, pole (≤15 km), line, minor_line, cable, substation, plant

Per-asset sub-points

- Distance (Haversine) · Asset kind · Voltage kV · Nearest overall/tower/pole/SS/line
- Interconnect ease: ≤0.25 km Easy · ≤2 km Easy · ≤12 km Moderate · >12 km Hard

Score bonuses (base: 0.2 km→10, 20 km→0 linear)

- Pole ≤ 350 m: +2 · Tower ≤ 2 km: +1.2 · Substation: +0.8 · Easy interconnect: +0.5 · Cap: 10
- No data: neutral score 5

6. Voltage suitability Weight 8%

Sub-point	Detail
Planning tiers	11, 33, 66, 110, 132, 220, 400, 765 kV
Suggested kV	Corridor → nearest tower → nearest line → nearest SS
Tagged + HV ≥33 kV nearby	Score ~9
Tagged only	Score ~7
No tags	Score 5 (neutral)

7. Connection distance Weight 8%

Sub-point	Formula / rule
Direct distance	Haversine to nearest grid asset (km)
Practical spur	Direct × 1.2 (screening factor)
≤ 0.3 km	Score 10
≥ 15 km	Score 0

8. Elevation Weight 8%

Elevation (m AMSL)	Score
20 – 800 m	9 (favourable)

< 5 m	3 (flood/coastal)
> 1800 m	4 (logistics)
Other / missing	7 neutral

9. Land cover hint Weight 8%

OSM Overpass 180 m: landuse, natural, water tags. Nominatim fallback.

Class	Score
Barren / open	9
Vegetated	5
Built-up	2
Water	0
Unknown	6

10. Soil / SBC Weight 8%

Path A — TAMS field geotech (within ~5 km)

Field	Use
Adopted SBC (T/m²)	≥20 → score 9 · ≥12 → score 7 · else 6
Design depth, CBR %, site code	Displayed in report

Path B — ISRIC SoilGrids 2.0 (~250 m pixel)

Depth bands	Properties
0–5, 5–15, 15–30, 30–60, 60–100, 100–200 cm	Clay%, sand%, silt%, bulk density, pH, coarse fragments, organic carbon
Texture	USDA class from avg 0–30 cm
Indicative SBC & CBR	From texture lookup (40–48% confidence)
Score	max(5, round(confidence% ÷ 10))

Not from GIS (FIELD TEST REQUIRED): SPT N · Atterberg · Soaked CBR · Cohesion · Resistivity · GWT

11. Wind exposure Weight 6%

Sub-point	Detail
Source	Open-Meteo Archive — last 90 days daily max wind @ 10 m
≤ 4 m/s	Score 10
≥ 12 m/s	Score 0

12. Corridor feasibility Weight 6%

Starts at score 8; deductions applied:

Check	Threshold	Penalty
Water proximity	< 150 m	–2
Settlement	< 100 m	–2
Steep slope	> 12°	–1.5
Very steep	> 18°	–1

C. Raw Signals at Analyze

Category	Sub-points stored
Location	Lat, lon, city/state (Nominatim)
Terrain	Elevation (m), slope (°), 5 DEM samples
Road	Distance (km), snap lat/lon
Water / Settlement	Distances (km), fallback flags
Climate	Mean wind (m/s)
Land cover	barren / vegetation / built / water / unknown
Grid	All assets; nearest tower/line/pole/SS; kV; interconnect ease; spur km
Soil	SoilGrids layers OR TAMS geotech record

D. Extra Screening (NOT in /10 score) — KML / Line Plan

Check	Sub-points
Voltage class	11–765 kV CEA-style bands
Span bands	Min · Ruling · Max span · ROW width (m)
Per pad T1...Tn	Chainage, lat/lon, verdict
Skip	Existing tower within ~35% of min span (40–120 m zone)
Shift	Closer than min span — suggested offset position
Review	Within 115% of ruling span
Suggest place	Spacing ≥ min span
Power take-off	Best pad → nearest SS; confidence %; interconnect links

E. Weight Summary

#	Factor	Weight
1	Terrain slope	22%
2	Road access	14%
3	Water / flood buffer	14%
4	Settlement clearance	10%
5	Power connectivity	10%
6	Voltage suitability	8%
7	Connection distance	8%
8	Elevation	8%
9	Land cover	8%
10	Soil / SBC	8%
11	Wind	6%
12	Corridor feasibility	6%

This document describes open-data screening logic in TAMS Tower Suitability. It is not utility approval, ROW certification, or foundation design. Lab values are only shown when entered in TAMS /geotech or marked FIELD TEST REQUIRED.